

Area of Parallelograms

Lesson 5-1

Name:

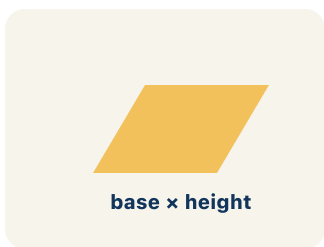
Date:

Class:

Key Vocabulary

Level 2 Standard

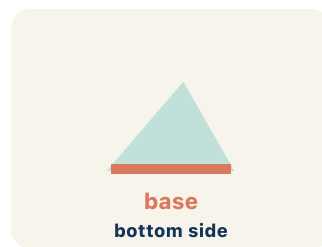
Picture first, then the word, then a plain-language meaning. Say each word out loud.



Think of a leaning door — the top and bottom are parallel, and the two sides are parallel, like a slanted rectangle

Parallelogram

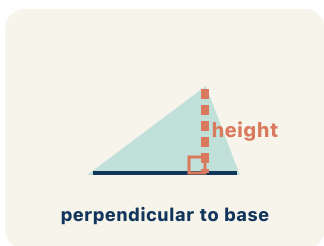
Write the definition:



If the bottom of a parallelogram is 10 cm, then $b = 10$ cm in the formula $A = b \times h$

Base

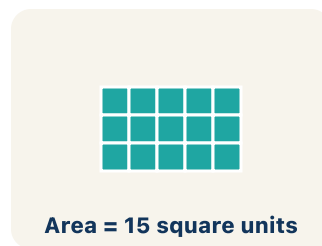
Write the definition:



A dashed vertical line from the top side straight down to the base, forming a 90° angle — NOT the slanted side

Height

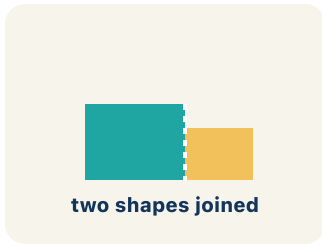
Write the definition:



A $3\text{ cm} \times 4\text{ cm}$ rectangle covers 12 square centimeters — imagine 12 tiny 1×1 squares inside it

Area

Write the definition:



An L-shaped room = a 10×8 rectangle joined to a 4×5 rectangle; total area = $80 + 20 = 100$ sq ft

Composite figure

Write the definition:

$$3x + 5$$

no equal sign

$A = b \times h$ means Area equals base times height; for a parallelogram with $b = 6$ and $h = 4$, $A = 24$

Formula

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can find the area of a parallelogram using $\text{base} \times \text{height}$.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Parallelogram

Base

Height

Area

Composite figure

Formula

1 A quadrilateral with two pairs of parallel sides is a _____.

2 Any side of a parallelogram used to calculate its area is the _____.

3 The perpendicular distance between the base and the opposite side is the _____.

4 The amount of flat space inside a two-dimensional figure is its _____.

5 A shape made of two or more basic shapes combined is a _____.

6 A math rule written with symbols, like $A = b \times h$, is a _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is the area of a parallelogram with base 10 cm and height 7 cm?

- A. 70 sq cm
- B. 34 sq cm
- C. 17 sq cm
- D. 70 cm

Step 1 $A = b \times h = 10 \times 7 = 70$ square centimeters.

 **Answer:** A. 70 sq cm

 We do — together

Solve this one with your class using the same steps.

Problem: A parallelogram has an area of 54 sq ft and a base of 9 ft. What is the height?

- A. 6 ft
- B. 45 ft
- C. 63 ft
- D. 27 ft

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: What is the area of a parallelogram with base 9 cm and height 5 cm?

- A. 45 sq cm
- B. 14 sq cm
- C. 22.5 sq cm
- D. 90 sq cm

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. A blueprint parallelogram has a base of 8 ft, a height of 5 ft, and a slanted side of 7 ft. What is its area?

- A. 40 sq ft
- B. 56 sq ft
- C. 35 sq ft
- D. 26 sq ft

Show your work:

2. Two blueprints share a base of 10 ft. Blueprint M has height 6 ft; Blueprint N is more tilted with the same height of 6 ft but a longer slanted side of 9 ft. How do their areas compare?

- A. Equal: both are $10 \times 6 = 60$ sq ft because area depends on base and height, not the slant
- B. Blueprint N is larger because its slanted side is 9 ft
- C. Blueprint M is larger because it is less tilted
- D. You cannot compare them without the slanted side of M

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A parallelogram has a base of 12 cm. If you double the height, the area doubles too. Explain why this happens using the formula, and give a specific example with numbers.

Sentence starter: When the height doubles from ____ to ____, the area changes from ____ to ____ because in the formula $A = b \times h$, the base stays at ____ and ____.

Show your work:

Reflect — Exit Ticket

A parallelogram has a base of 13 inches and a height of 5 inches. What is its area?

- A. 65 sq in
- B. 36 sq in
- C. 18 sq in
- D. 65 in

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Parallelogram
2. **Notes 2:** Base
3. **Notes 3:** Height
4. **Notes 4:** Area
5. **Notes 5:** Composite figure
6. **Notes 6:** Formula
7. **We Try (notes):** A. $6 \text{ ft} - A = b \times h \rightarrow 54 = 9 \times h \rightarrow h = 54 \div 9 = 6 \text{ ft}.$
8. **You Try (notes):** A. $45 \text{ sq cm} - A = b \times h = 9 \times 5 = 45 \text{ square centimeters}.$
9. **Try It 1:** A. $40 \text{ sq ft} - \text{Area} = \text{base} \times \text{height} = 8 \times 5 = 40 \text{ sq ft}.$ *The 7 ft slanted side is not used; only the perpendicular height belongs in the formula.*
10. **Try It 2:** A. Equal: both are $10 \times 6 = 60 \text{ sq ft}$ because area depends on base and height, not the slant — $\text{Area} = \text{base} \times \text{height}.$ *Both have base 10 ft and height 6 ft, so both areas are $10 \times 6 = 60 \text{ sq ft}.$ Tilting the shape changes the slanted side but not the perpendicular height, so the area stays the same.*
11. **Exit Ticket:** A. $65 \text{ sq in} - A = b \times h = 13 \times 5 = 65 \text{ square inches}.$